Linux Disk Partitioning

Dividing a disk into partitions with fdisk, formatting with a filesystem, and mounting - with examples, a flow diagram, command chart and interview Q&A.

ROOT = changes the system or reads protected data; run it from a root shell. **ANY USER** = read-only, safe for everyone.

How to read this: partitioning, formatting and mounting all change the system, so every working example here is ROOT. Read-only listing commands are safe for any user. Replace /dev/sdb and sizes with your real disk and values - running these against the wrong disk destroys data.

1. What disk partitioning is

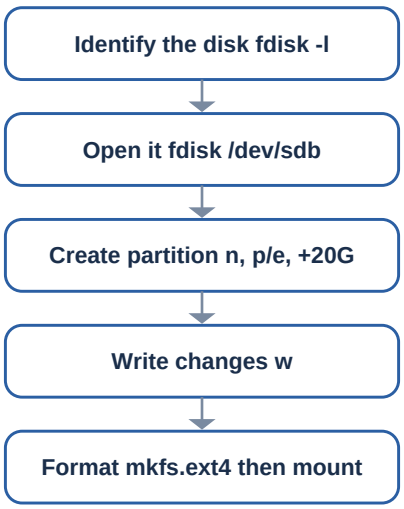
Disk partitioning divides one physical storage device into multiple logical sections called **partitions**. Each partition behaves like its own disk: it can hold a different filesystem, a different operating system, or a separate area of data.

Why partition at all? It gives better **data management** - system files stay separate from user data, so a full home partition cannot crash the OS - and it lets one machine **run multiple operating systems** side by side, each on its own partition.

Two partition styles you will meet: the older **MBR** scheme allows up to four **primary** partitions, so it invented the **extended** partition - a container that holds further **logical** partitions. You never format the extended container itself - you create logical partitions inside it and format those. The newer **GPT** scheme drops this primary/extended split and allows many more partitions and larger disks. When fdisk offers you primary (p) or extended (e), that choice comes from MBR.

2. The partitioning workflow

Creating a usable partition is always the same five steps:



Step 1 - Identify the disk

```
# ROOT - lists all disks and their partitions
fdisk -l
```

Find the disk you want (for example /dev/sdb) by its size. **Never** guess - partitioning the wrong disk erases it.

Step 2 - Open the disk in fdisk

```
# ROOT
fdisk /dev/sdb
```

fdisk works inside its own command mode: you type single letters at its prompt. Useful keys: `m` shows the menu, `p` prints the current partition table, `q` quits without saving.

Step 3 - Create the new partition

Type `n` for a new partition, then answer its questions:

Type - choose `p` (primary) or `e` (extended). **Partition number** - the default is usually fine. **Starting sector** - the default is typically safe, it begins right after existing data. **Size** - give it as `+sizeG`, for example `+20G` for a 20 GiB partition (fdisk's G means 1024 MiB - binary units, slightly more than the 20 GB a disk maker advertises).

```
Command (m for help): n
Partition type: p
Partition number: 1
First sector: <press Enter for the default>
Last sector, +/-sectors or +/-size{K,M,G,T,P}: +20G
```

Step 4 - Write the changes

Nothing has changed on the disk yet - fdisk only stages your choices. Type `p` to review, then `w` to write the table to disk. Until `w`, `q` abandons everything safely.

Step 5 - Format and mount

A new partition is raw space; give it a **filesystem** (for example ext4) and then attach it to a directory with mount:

```
# ROOT - format the new partition (replace sdb1 with your partition)
mkfs.ext4 /dev/sdb1
# ROOT - create the mount point if it does not exist yet
mkdir -p /mnt/data
# ROOT - attach the partition there
mount /dev/sdb1 /mnt/data
```

A mount done by hand lasts until reboot. To mount automatically every boot, add the partition to `/etc/fstab`. Check your mounts any time with `lsblk` or `df -h`.

Command chart

Command	What it does	Example
`fdisk -l`	List all disks and partitions (ROOT)	`fdisk -l`
`fdisk /dev/sdb`	Open a disk in fdisk command mode (ROOT)	`fdisk /dev/sdb`
`n, p/e, +20G`	Inside fdisk: new partition, type, size	`+20G = 20 GiB`
`p / w / q`	Inside fdisk: print table / write / quit unsaved	`w writes changes`
`mkfs.ext4`	Format a partition with ext4 (ROOT)	`mkfs.ext4 /dev/sdb1`
`mkdir / mount`	Create the mount point, then attach the partition (ROOT)	`mkdir -p /mnt/data; mount /dev/sdb1 /mnt/data`
`umount`	Detach a mounted partition (ROOT)	`umount /mnt/data`
`lsblk`	Show disks, partitions and mount points (ANY USER)	`lsblk`

Interview Q&A

Q. What is a partition?

A logical section of a physical disk. The disk is divided so each section can hold its own filesystem or operating system.

Q. Why partition a disk?

Better data management (separating system and user data so one full area cannot crash the OS) and the ability to run multiple operating systems on one machine.

Q. Primary vs extended partition?

Under the MBR scheme only four primary partitions are allowed; an extended partition is a container that holds additional logical partitions to get past that limit. The container itself is never formatted - the logical partitions inside it are.

Q. Which fdisk keys create and save a partition?

n starts a new partition, p or e picks primary or extended, you accept the default number and start, give a size like +20G, and w writes the changes. Nothing changes until w.

Q. What is the difference between partitioning, formatting and mounting?

Partitioning carves out the space (fdisk), formatting builds a filesystem on it (mkfs.ext4), and mounting attaches it to a directory so you can use it (mount).

Q. A manually mounted partition disappears after reboot. Why?

Hand mounts are not persistent. To mount at every boot, add an entry to /etc/fstab.

Sources: class slides (Disk Partitioning) and the Linux man pages (man fdisk, man mkfs, man mount, man fstab).